Unit-interval retracts of quasicontinuous domains: a counterexample

Abstract

We construct a quasicontinuous complete lattice P which is neither continuous nor quasialgebraic and in which every chain is at most countable. Consequently, P contains no sub-dcpo order-isomorphic to the unit interval and does not admit $[0, 1]$ as a Scott-continuous retract. This gives negative answers to two problems posed by Xu. Nevertheless, there is an explicit Scott-continuous surjection from P onto $[0, 1]$. The principal technical result is an exact description of the finite-set way-below relation. It explains why finite approximation yields quasicontinuity whereas one-point approximation fails. We also characterize when a Scott-continuous surjection onto $[0, 1]$ has a section by means of a compatible increasing family indexed by the dyadic rationals.

Keywords: quasicontinuous domain, quasialgebraic domain, Scott-continuous retract, unit interval, finite-set way-below relation
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1. Introduction

Jia, Li and Luan proved that every nonalgebraic continuous domain admits the unit interval $[0, 1]$ as a Scott-continuous retract [3]. Xu subsequently proved that every nonquasialgebraic quasicontinuous domain admits a surjective monotone Lawson-continuous map onto $[0, 1]$ [5]. Motivated by these results, Xu posed the following two problems in the contribution "Two problems related to quasicontinuous domains" to the collection *Open Problems in Domain Theory*, prepared for the Domain Theory Seminar at TYMC in December 2025 [4, p. 33].

Problem 1.1. *Let P be a quasicontinuous domain which is not quasialgebraic. Must $[0, 1]$ be a Scott-continuous retract of P?*

Problem 1.2. *Let P be a quasicontinuous domain which is not quasialgebraic. Must P contain a sub-dcpo order-isomorphic to $[0, 1]$?*

Our main result gives negative answers to both questions.

Theorem 1.3. *There exists a complete lattice P with the following properties:*

1. *P is quasicontinuous;*
2. *P is not continuous;*
3. *P is not quasialgebraic;*
4. *every chain in P is at most countable;*
5. *there is a Scott-continuous surjection $q : P \to [0, 1]$.*

Consequently, P contains no sub-dcpo order-isomorphic to $[0, 1]$, and $[0, 1]$ is not a Scott-continuous retract of P.

The example is a sparse upper hyperspace over $X = [0, 1]$. Its elements are the empty set, all singleton subsets of X, and a countable skeleton of nondegenerate closed intervals, ordered by reverse inclusion. The resulting poset is a complete lattice. Its uncountability is concentrated in the singleton layer, which is an antichain, and hence every chain is at most countable.

The central point is the approximation structure. For a nonempty lattice element K and a nonempty finite family $\mathcal{F}$ of lattice elements, we prove the exact formula

$$\mathcal{F} \ll K \quad \Longleftrightarrow \quad \begin{cases} K \subseteq A_0 \text{ for some } A_0 \in \mathcal{F}, \\ K \subseteq \bigcup_{A \in \mathcal{F}} \operatorname{int}_X A. \end{cases}$$

Finite approximants may cover the boundary points of a nondegenerate interval by different members of $\mathcal{F}$, and this yields quasicontinuity. By contrast, a one-point approximant must contain the entire interval in its interior. The interval skeleton is deliberately too sparse to provide enough such approximants at a particular interval, and continuity fails there. Finally, if $\mathcal{F} \ll \mathcal{F}$, then the trace of the Scott-open finitely generated upper set $\uparrow_P \mathcal{F}$ on the singleton layer is clopen in $[0, 1]$. Connectedness therefore prevents these upper sets from separating distinct singletons, which yields failure of quasialgebraicity.

The construction and its complete-lattice structure are developed in Section 3. The finite-set way-below relation is computed in Section 4. Quasicontinuity, failure of continuity, and failure of quasialgebraicity are proved independently in Section 5. The proof of Theorem 1.3, including the explicit Scott-continuous surjection, is completed in Section 6. A short final section gives an exact criterion for a Scott-continuous surjection onto $[0, 1]$ to have a section, expressed through an increasing family indexed by the dyadic rationals.

2. Preliminaries

Let P be a poset. A nonempty subset $\mathcal{D} \subseteq P$ is *directed* if every finite subset of $\mathcal{D}$ has an upper bound in $\mathcal{D}$. A *dcpo* is a poset in which every directed subset has a supremum. The Scott topology $\sigma(P)$ on a dcpo P consists of the upper subsets $U \subseteq P$ such that

$$\bigvee \mathcal{D} \in U \quad \Longrightarrow \quad \mathcal{D} \cap U \neq \varnothing$$

for every directed $\mathcal{D} \subseteq P$. A map between dcpos is Scott continuous if and only if it is monotone and preserves directed suprema [1].

For a subset $\mathcal{A} \subseteq P$, write

$$\uparrow_P \mathcal{A} = \{y \in P : a \leq y \text{ for some } a \in \mathcal{A}\}.$$

If $\mathcal{A} = \{a\}$, we write $\uparrow_P a$. For subsets $\mathcal{A}, C \subseteq P$, write $\mathcal{A} \ll C$ if, for every directed set $\mathcal{D} \subseteq P$,

$$\bigvee \mathcal{D} \in \uparrow_P C \quad \Longrightarrow \quad \mathcal{D} \cap \uparrow_P \mathcal{A} \neq \varnothing. \tag{1}$$

When $\mathcal{A} = \{a\}$ and $C = \{c\}$, this is the usual way-below relation and is denoted by $a \ll c$. Let $P^{(<\omega)}$ be the set of all nonempty finite subsets of P, and define

$$\mathcal{W}(p) = \{\mathcal{F} \in P^{(<\omega)} : \mathcal{F} \ll \{p\}\}.$$

We abbreviate $\mathcal{F} \ll \{p\}$ to $\mathcal{F} \ll p$.

For $p \in P$, put

$$\Downarrow p = \{u \in P : u \ll p\}.$$

A dcpo P is *continuous* if $\Downarrow p$ is directed and $p = \bigvee \Downarrow p$ for every $p \in P$. It is *quasicontinuous* if, for every $p \in P$, the family

$$\{\uparrow_P \mathcal{F} : \mathcal{F} \in \mathcal{W}(p)\}$$

is filtered under inclusion and

$$\uparrow_P p = \bigcap_{\mathcal{F} \in \mathcal{W}(p)} \uparrow_P \mathcal{F}. \tag{2}$$

It is *quasialgebraic* if, for every $p \in P$, the family

$$\{\uparrow_P \mathcal{F} : \mathcal{F} \in P^{(<\omega)},\ p \in \uparrow_P \mathcal{F},\ \mathcal{F} \ll \mathcal{F}\}$$

is filtered and has intersection $\uparrow_P p$. These formulations are standard; see [1, 2, 4]. Following Xu's terminology, a "quasicontinuous domain" in this paper means a quasicontinuous dcpo, and similarly for "quasialgebraic domain".

We shall use the following elementary observation repeatedly.

Lemma 2.1. *For every $\mathcal{F} \in P^{(<\omega)}$,*

$$\mathcal{F} \ll \mathcal{F} \quad \Longleftrightarrow \quad \uparrow_P \mathcal{F} \text{ is Scott open.}$$

Proof. The set $\uparrow_P \mathcal{F}$ is an upper set. By (1), the relation $\mathcal{F} \ll \mathcal{F}$ says exactly that $\uparrow_P \mathcal{F}$ is inaccessible by directed suprema. This is the Scott-openness condition. □

A subset $C \subseteq P$ is a *sub-dcpo* if it is a dcpo with the inherited order and, for every directed $\mathcal{D} \subseteq C$, the supremum of $\mathcal{D}$ in C agrees with its supremum in P. A dcpo Q is a *Scott-continuous retract* of a dcpo P if there are Scott-continuous maps

$$r : P \longrightarrow Q, \qquad s : Q \longrightarrow P$$

such that $r \circ s = \mathrm{id}_Q$. The map s is a section of r.

We use the following notational convention in the construction below. Lower-case letters x, y, a, b, s, t, r, d and e denote points of X or scalar parameters. Capital Roman letters A, B, C, D, E, K, L and S denote elements of P, and hence subsets of X. Calligraphic letters $\mathcal{A}$ and $\mathcal{D}$ denote arbitrary and directed subfamilies of P, respectively, while $\mathcal{F}$ and $\mathcal{G}$ denote nonempty finite subfamilies. The symbol $\mathcal{B}$ is reserved for the interval skeleton.

3. The sparse interval lattice

3.1. A countable interval skeleton

Let $X = [0, 1]$ with its usual metric and relative topology. For $n \geq 1$ and $0 \leq k \leq 2^n$, define

$$J_{n,k} = X \cap \left[\frac{k-1}{2^n}, \frac{k+1}{2^n} \right]. \tag{3}$$

Let $\mathcal{B}$ be the family of all nondegenerate finite intersections of the intervals in (3):

$$\mathcal{B} = \left\{ \bigcap_{i=1}^{r} J_{n_i,k_i} : r \geq 1, \quad \operatorname{diam}\left(\bigcap_{i=1}^{r} J_{n_i,k_i} \right) > 0 \right\}. \tag{4}$$

Notice that $X = J_{1,1}$ belongs to $\mathcal{B}$.

Lemma 3.1. *The family $\mathcal{B}$ satisfies the following properties.*

1. *Every member of $\mathcal{B}$ is a nondegenerate closed interval in X.*
2. *The family $\mathcal{B}$ is countable.*
3. *For every $\varepsilon > 0$, the set*

$$\{B \in \mathcal{B} : \operatorname{diam}(B) \geq \varepsilon\}$$

is finite.

Proof. Each $J_{n,k}$ is a closed interval in X. A finite intersection of closed intervals is either empty, a singleton, or a closed interval. The definition of $\mathcal{B}$ retains exactly the intersections of positive diameter, proving (1).

The index set

$$\{(n, k) : n \geq 1,\ 0 \leq k \leq 2^n\}$$

is countable. The collection of its nonempty finite subsets is countable, and each member of $\mathcal{B}$ is represented by one such finite subset. Hence $\mathcal{B}$ is countable.

For every n and k,

$$\operatorname{diam}(J_{n,k}) \leq 2^{1-n}. \tag{5}$$

Fix $\varepsilon > 0$. If $\operatorname{diam}(J_{n,k}) \geq \varepsilon$, then $2^{1-n} \geq \varepsilon$, so only finitely many values of n are possible; for each such n, only finitely many values of k occur. Thus there are only finitely many basic intervals $J_{n,k}$ of diameter at least ε.

Now let

$$B = J_{n_1,k_1} \cap \cdots \cap J_{n_r,k_r} \in \mathcal{B}$$

and assume $\operatorname{diam}(B) \geq \varepsilon$. Since $B \subseteq J_{n_i,k_i}$ for every i, each interval occurring in the intersection has diameter at least ε. Consequently, B is an intersection of members of a fixed finite family of basic intervals. A finite family has only finitely many distinct intersections. This proves (3). □

Lemma 3.2. *For every $x \in X$ and every open neighbourhood O of x in X, there exists $B \in \mathcal{B}$ such that*

$$x \in \operatorname{int}_X B \subseteq B \subseteq O.$$

Proof. Choose $\delta > 0$ such that

$$X \cap (x - \delta, x + \delta) \subseteq O.$$

For each sufficiently large n, choose an integer k with $0 \le k \le 2^n$ nearest to $2^n x$. Then

$$\left| x - \frac{k}{2^n} \right| \le 2^{-n-1} < 2^{-n}. \tag{6}$$

The strict inequality in (6) shows that x belongs to the relative interior of $J_{n,k}$. If $y \in J_{n,k}$, then

$$|y - x| \le \left| y - \frac{k}{2^n} \right| + \left| \frac{k}{2^n} - x \right| \le 2^{-n} + 2^{-n-1}.$$

For sufficiently large n, the last quantity is smaller than δ. Hence $J_{n,k} \subseteq O$. Taking $B = J_{n,k}$ proves the assertion. □

For later use, fix, for every $x \in X$, a sequence $(N_m(x))_{m \ge 1}$ in $\mathcal{B}$ such that

$$x \in \operatorname{int}_X N_m(x), \qquad N_{m+1}(x) \subseteq \operatorname{int}_X N_m(x), \qquad \operatorname{diam}(N_m(x)) \longrightarrow 0. \tag{7}$$

To construct it, first apply Theorem 3.2 with $O = X$ and choose $N_1(x)$. Once $N_m(x)$ has been chosen, apply the same lemma to the open neighbourhood

$$\operatorname{int}_X N_m(x) \cap \{y \in X : |y - x| < 1/(m+1)\}$$

to obtain $N_{m+1}(x)$. Then

$$\operatorname{diam}(N_{m+1}(x)) \le \frac{2}{m+1},$$

so the last condition in (7) holds. The sequence is decreasing under inclusion and contains x in every term. Consequently,

$$\bigcap_{m \ge 1} N_m(x) = \{x\}. \tag{8}$$

Indeed, a point belonging to all $N_m(x)$ has distance at most $\operatorname{diam}(N_m(x))$ from x for every m, and hence must equal x.

3.2. The complete lattice

Define

$$P = \mathcal{B} \cup \{\{x\} : x \in X\} \cup \{\varnothing\}, \tag{9}$$

and order P by reverse inclusion:

$$A \le_P B \quad \Longleftrightarrow \quad A \supseteq B. \tag{10}$$

We shall write

$$\bot := X, \qquad \top := \varnothing. \tag{11}$$

Thus $\bot$ and $\top$ are, respectively, the least and greatest elements of P. The notation $\top$ will be used whenever the empty subset of X is regarded as an order-theoretic element of P; the symbol $\varnothing$ will remain available for an empty family or an empty topological set.

Proposition 3.3. *The poset P is a complete lattice. For every family $\mathcal{A} \subseteq P$,*

$$\bigvee_P \mathcal{A} = \bigcap \mathcal{A}, \tag{12}$$

where the intersection of the empty family is $X = \bot$.

Proof. We first prove that P is closed under arbitrary intersections. If $\mathcal{A} = \varnothing$, then

$$\bigcap \mathcal{A} = X = J_{1,1} \in \mathcal{B}.$$

Assume that $\mathcal{A}$ is nonempty and put

$$S = \bigcap \mathcal{A}.$$

Every member of P is a closed convex subset of X. Hence S is a closed convex subset of X and is therefore either empty, a singleton, or a closed interval. If $S = \varnothing$, then $S = \top \in P$. If S is a singleton, then $S \in P$ by definition.

It remains to consider the case

$$\delta := \operatorname{diam}(S) > 0.$$

No member of $\mathcal{A}$ can then be $\top$ or a singleton. Thus every member of $\mathcal{A}$ belongs to $\mathcal{B}$. Moreover, each $A \in \mathcal{A}$ contains S, and hence

$$\operatorname{diam}(A) \geq \operatorname{diam}(S) = \delta.$$

By Theorem 3.1(3), only finitely many members of $\mathcal{B}$ have diameter at least δ. Therefore $\mathcal{A}$ has only finitely many distinct members. It follows that S is an intersection of finitely many members of $\mathcal{B}$, and hence a finite intersection of basic intervals $J_{n,k}$. Since $\operatorname{diam}(S) > 0$, definition (4) yields $S \in \mathcal{B}$.

Thus P is closed under arbitrary intersections. Under reverse inclusion, $\bigcap \mathcal{A}$ is the least upper bound of $\mathcal{A}$: every $A \in \mathcal{A}$ contains $\bigcap \mathcal{A}$, and any upper bound of $\mathcal{A}$ is contained in every $A \in \mathcal{A}$ and therefore in their intersection. This proves (12). Since all suprema exist, all infima exist as suprema of sets of lower bounds, and P is a complete lattice. □

The following compactness lemma is the main tool for passing from a filtered intersection to one member of a directed family.

Lemma 3.4. *Let $\mathcal{D} \subseteq P$ be directed and put*

$$S = \bigcap \mathcal{D}.$$

If $O \subseteq X$ is open and $S \subseteq O$, then there exists $D_0 \in \mathcal{D}$ such that $D_0 \subseteq O$.

Proof. Assume, to the contrary, that no member of $\mathcal{D}$ is contained in O. Then

$$D \cap (X \setminus O) \neq \varnothing \qquad (D \in \mathcal{D}).$$

Each set $D \cap (X \setminus O)$ is compact, being a closed subset of the compact space X.

We verify the finite-intersection property. Let $D_1, \ldots, D_r \in \mathcal{D}$. Directedness with respect to the reverse-inclusion order gives $E \in \mathcal{D}$ such that

$$D_i \leq_P E \qquad (1 \leq i \leq r).$$

By (10),

$$E \subseteq D_1 \cap \cdots \cap D_r.$$

Since $E \not\subseteq O$, choose a point of $E \cap (X \setminus O)$. This point belongs to

$$\bigcap_{i=1}^{r} (D_i \cap (X \setminus O)).$$

Thus the compact family

$$\{D \cap (X \setminus O) : D \in \mathcal{D}\}$$

has the finite-intersection property. Compactness of X gives

$$\varnothing \neq \bigcap_{D \in \mathcal{D}} (D \cap (X \setminus O)) = S \cap (X \setminus O),$$

contrary to $S \subseteq O$. Hence some $D_0 \in \mathcal{D}$ is contained in O. □

Corollary 3.5. *If $\mathcal{D} \subseteq P$ is directed and $\bigcap \mathcal{D} = \top$, then $\top \in \mathcal{D}$.*

Proof. Since $\top = \varnothing$, apply Theorem 3.4 to the open set $O = \varnothing$. □

4. The finite-set way-below relation

For every $\mathcal{F} \in P^{(<\omega)}$, reverse inclusion gives the concrete description

$$C \in{\uparrow_P} \mathcal{F} \quad \Longleftrightarrow \quad C \subseteq A \text{ for some } A \in \mathcal{F}. \tag{13}$$

We now compute the finite-set way-below relation exactly. This is the main technical result of the paper.

Theorem 4.1. *Let $K \in P \setminus \{\top\}$ and let $\mathcal{F} \in P^{(<\omega)}$. Then the following conditions are equivalent:*

1. *$\mathcal{F} \ll K$;*
2. *both*

$$K \subseteq A_0 \quad \text{for some } A_0 \in \mathcal{F}, \tag{14}$$

$$K \subseteq \bigcup_{A \in \mathcal{F}} \operatorname{int}_X A \tag{15}$$

hold.

For $K = \top$, every $\mathcal{F} \in P^{(<\omega)}$ satisfies $\mathcal{F} \ll \top$.

Proof. Assume first that $K \neq \top$ and $\mathcal{F} \ll K$.

Apply the definition of $\mathcal{F} \ll K$ to the one-point directed set $\{K\}$. Its supremum is K, and $K \in{\uparrow_P} K$. Hence

$$\{K\} \cap {\uparrow_P} \mathcal{F} \neq \varnothing.$$

Thus $K \in{\uparrow_P} \mathcal{F}$, and (13) yields an element $A_0 \in \mathcal{F}$ such that $K \subseteq A_0$. This proves (14).

To prove (15), fix $x \in K$. The sequence

$$N_1(x) \supseteq N_2(x) \supseteq \cdots$$

is increasing in the order of P. By Theorem 3.3 and (8),

$$\bigvee_{m \geq 1} N_m(x) = \bigcap_{m \geq 1} N_m(x) = \{x\}.$$

Since $x \in K$, one has $\{x\} \subseteq K$, and therefore $\{x\} \in{\uparrow_P} K$. The relation $\mathcal{F} \ll K$ gives an index m such that

$$N_m(x) \in{\uparrow_P} \mathcal{F}.$$

By (13), $N_m(x) \subseteq A$ for some $A \in \mathcal{F}$. Since $x \in \operatorname{int}_X N_m(x)$, it follows that $x \in \operatorname{int}_X A$. As $x \in K$ was arbitrary, (15) follows.

Conversely, assume (14) and (15). Let $\mathcal{D} \subseteq P$ be directed and suppose that

$$\bigvee_P \mathcal{D} \in{\uparrow_P} K.$$

By Theorem 3.3,

$$S := \bigvee_P \mathcal{D} = \bigcap \mathcal{D}.$$

The condition $S \in{\uparrow_P} K$ is equivalent, under reverse inclusion, to

$$S \subseteq K. \tag{16}$$

The closed convex subset S of X is empty, a singleton, or has positive diameter. We treat the three cases separately.

If $S = \top$, then Theorem 3.5 gives $\top \in \mathcal{D}$. Since $\top = \varnothing$ is contained in every $A \in \mathcal{F}$, one has $\top \in{\uparrow_P} \mathcal{F}$. Hence $\mathcal{D} \cap {\uparrow_P} \mathcal{F} \neq \varnothing$.

Suppose that $S = \{x\}$. By (16), $x \in K$. Condition (15) gives $A \in \mathcal{F}$ with $x \in \operatorname{int}_X A$. Thus

$$S = \{x\} \subseteq \operatorname{int}_X A.$$

By filtered compactness, there is $D_0 \in \mathcal{D}$ such that

$$D_0 \subseteq \operatorname{int}_X A \subseteq A.$$

Consequently, $D_0 \in \uparrow_P \mathcal{F}$.

Finally, suppose that $\operatorname{diam}(S) > 0$. Every $D \in \mathcal{D}$ contains S, and hence

$$\operatorname{diam}(D) \geq \operatorname{diam}(S) > 0.$$

Thus every member of $\mathcal{D}$ belongs to $\mathcal{B}$. By Theorem 3.1(3), only finitely many members of $\mathcal{B}$ have diameter at least $\operatorname{diam}(S)$. Therefore $\mathcal{D}$ is finite. A finite directed set has a greatest element; denote it by D_*. Since the order is reverse inclusion, D_* is contained in every member of $\mathcal{D}$, and hence

$$D_* = \bigcap \mathcal{D} = S.$$

Choose $A_0 \in \mathcal{F}$ as in (14). Then

$$D_* = S \subseteq K \subseteq A_0,$$

so $D_* \in \uparrow_P \mathcal{F}$.

In every case, $\mathcal{D} \cap \uparrow_P \mathcal{F} \neq \varnothing$. Hence $\mathcal{F} \ll K$.

It remains to consider $K = \top$. Let $\mathcal{F} \in P^{(<\omega)}$, and let $\mathcal{D} \subseteq P$ be directed with

$$\bigvee_P \mathcal{D} \in \uparrow_P \top.$$

Since $\top$ is the greatest element, $\uparrow_P \top = \{\top\}$, and therefore $\bigvee_P \mathcal{D} = \top$. By Theorem 3.5, $\top \in \mathcal{D}$. Also $\top \in \uparrow_P \mathcal{F}$, because $\top = \varnothing$ is contained in every member of $\mathcal{F}$. Thus $\mathcal{F} \ll \top$. □

Corollary 4.2. *Let $K \in P \setminus \{\top\}$ and $A \in P$. Then*

$$A \ll K \quad \Longleftrightarrow \quad K \subseteq \operatorname{int}_X A. \tag{17}$$

Proof. Apply Theorem 4.1 to the singleton family $\mathcal{F} = \{A\}$. Condition (15) is exactly $K \subseteq \operatorname{int}_X A$, and it implies (14). □

5. Approximation properties

5.1. Quasicontinuity

For $m \geq 1$, define a map

$$\mathcal{F}_m : P \longrightarrow P^{(<\omega)}$$

by

$$\mathcal{F}_m(K) = \begin{cases} \{N_m(x)\}, & K = \{x\}, \\ \{K, N_m(a), N_m(b)\}, & K = [a, b] \in \mathcal{B}, \\ \{\top\}, & K = \top. \end{cases} \tag{18}$$

Lemma 5.1. *For every $K \in P$ and every $m \geq 1$,*

$$\mathcal{F}_m(K) \ll K.$$

Proof. Suppose first that $K = \{x\}$. The unique member $N_m(x)$ of $\mathcal{F}_m(K)$ contains K, and $x \in \operatorname{int}_X N_m(x)$. Hence both conditions of Theorem 4.1 hold.

Let $K = [a, b] \in \mathcal{B}$. Since $K \in \mathcal{F}_m(K)$, condition (14) holds. Moreover,

$$K \setminus \operatorname{int}_X K \subseteq \{a, b\},$$

where the inclusion is understood in the relative topology of X. By construction,

$$a \in \operatorname{int}_X N_m(a), \qquad b \in \operatorname{int}_X N_m(b).$$

Therefore

$$K \subseteq \operatorname{int}_X K \cup \operatorname{int}_X N_m(a) \cup \operatorname{int}_X N_m(b),$$

and Theorem 4.1 gives $\mathcal{F}_m(K) \ll K$.

If $K = \top$, the last assertion of Theorem 4.1 gives $\{\top\} \ll \top$. □

Lemma 5.2. *Let $K \in P$ and let $\mathcal{G} \in P^{(<\omega)}$ satisfy $\mathcal{G} \ll K$. Then*

$$\uparrow_P \mathcal{F}_m(K) \subseteq \uparrow_P \mathcal{G} \tag{19}$$

for all sufficiently large m.

Proof. Suppose first that $K = \{x\}$. By (15), some $A \in \mathcal{G}$ satisfies $x \in \operatorname{int}_X A$. Choose an open neighbourhood V of x with $V \subseteq A$. Since $x \in N_m(x)$ and $\operatorname{diam}(N_m(x)) \to 0$, one has

$$N_m(x) \subseteq V \subseteq A$$

for all sufficiently large m. If $C \in \uparrow_P \mathcal{F}_m(K)$, then $C \subseteq N_m(x) \subseteq A$, and hence $C \in \uparrow_P \mathcal{G}$.

Let $K = [a, b] \in \mathcal{B}$. By Theorem 4.1, choose $A_0, A_a, A_b \in \mathcal{G}$ such that

$$K \subseteq A_0, \qquad a \in \operatorname{int}_X A_a, \qquad b \in \operatorname{int}_X A_b.$$

The three chosen members need not be distinct. Since the neighbourhoods $N_m(a)$ and $N_m(b)$ shrink to their centres,

$$N_m(a) \subseteq A_a, \qquad N_m(b) \subseteq A_b$$

for all sufficiently large m. Thus every member of $\mathcal{F}_m(K)$ is contained in a member of $\mathcal{G}$. Formula (13) then gives (19).

Finally, let $K = \top$. Then

$$\uparrow_P \mathcal{F}_m(\top) = \uparrow_P \{\top\} = \{\top\}.$$

Because $\top$ is contained in every member of the nonempty family $\mathcal{G}$, one has $\top \in \uparrow_P \mathcal{G}$. Hence (19) holds for every m. □

Lemma 5.3. *For every $K \in P$,*

$$\bigcap_{m \geq 1} \uparrow_P \mathcal{F}_m(K) = \uparrow_P K. \tag{20}$$

Proof. Let $K = \{x\}$. By (13), an element $C \in P$ belongs to every $\uparrow_P \mathcal{F}_m(K)$ if and only if

$$C \subseteq N_m(x) \qquad (m \geq 1).$$

By (8), this is equivalent to $C \subseteq \{x\}$, which in turn is equivalent to $C \in \uparrow_P \{x\}$.

Let $K = [a, b] \in \mathcal{B}$. Since $K \in \mathcal{F}_m(K)$, one has

$$\uparrow_P K \subseteq \bigcap_{m \geq 1} \uparrow_P \mathcal{F}_m(K).$$

For the reverse inclusion, let C belong to the intersection and suppose, for a contradiction, that $C \not\subseteq K$. For each m, formula (13) gives

$$C \subseteq K, \quad \text{or} \quad C \subseteq N_m(a), \quad \text{or} \quad C \subseteq N_m(b).$$

The first alternative is excluded. Hence one of the last two alternatives occurs for infinitely many m.

Suppose, for example, that $C \subseteq N_m(a)$ for infinitely many m. Given $r \geq 1$, choose such an $m \geq r$. Since the sequence is decreasing,

$$C \subseteq N_m(a) \subseteq N_r(a).$$

Thus $C \subseteq N_r(a)$ for every r, and (8) gives

$$C \subseteq \bigcap_{r \geq 1} N_r(a) = \{a\} \subseteq K,$$

a contradiction. The argument at b is identical. Therefore $C \subseteq K$, proving the reverse inclusion.

If $K = \top$, then $\mathcal{F}_m(K) = \{\top\}$ for every m, and both sides of (20) equal $\{\top\}$. □

Theorem 5.4. *The complete lattice P is quasicontinuous.*

Proof. Fix $K \in P$. By Theorem 5.1, $\mathcal{F}_m(K) \in \mathcal{W}(K)$ for every m.

We first verify filteredness. Let $\mathcal{G}_1, \mathcal{G}_2 \in \mathcal{W}(K)$. By Theorem 5.2, there are integers m_1, m_2 such that

$$\uparrow_P \mathcal{F}_m(K) \subseteq \uparrow_P \mathcal{G}_i \qquad (m \geq m_i,\ i = 1, 2).$$

Choose $m \geq \max\{m_1, m_2\}$. Then

$$\uparrow_P \mathcal{F}_m(K) \subseteq \uparrow_P \mathcal{G}_1 \cap \uparrow_P \mathcal{G}_2,$$

and $\mathcal{F}_m(K) \in \mathcal{W}(K)$. Hence the family

$$\{\uparrow_P \mathcal{G} : \mathcal{G} \in \mathcal{W}(K)\}$$

is filtered under inclusion.

For the intersection condition, let $\mathcal{G} \in \mathcal{W}(K)$. Applying $\mathcal{G} \ll K$ to the directed singleton $\{K\}$ gives $K \in \uparrow_P \mathcal{G}$. Since $\uparrow_P \mathcal{G}$ is an upper set,

$$\uparrow_P K \subseteq \uparrow_P \mathcal{G}.$$

Therefore

$$\uparrow_P K \subseteq \bigcap_{\mathcal{G} \in \mathcal{W}(K)} \uparrow_P \mathcal{G}. \tag{21}$$

Conversely, $\mathcal{F}_m(K) \in \mathcal{W}(K)$ for every m, so Theorem 5.3 gives

$$\bigcap_{\mathcal{G} \in \mathcal{W}(K)} \uparrow_P \mathcal{G} \subseteq \bigcap_{m \geq 1} \uparrow_P \mathcal{F}_m(K) = \uparrow_P K.$$

Together with (21), this is precisely (2). Thus P is quasicontinuous. □

5.2. *Failure of continuity*

We next isolate an element at which one-point approximation fails.

Lemma 5.5. *Let*

$$K_0 = J_{2,2} = \left[\frac{1}{4}, \frac{3}{4}\right].$$

Then

$$\downarrow_P K_0 = \{\bot, K_0\}. \tag{22}$$

Proof. Under reverse inclusion,

$$A \in \downarrow_P K_0 \quad \Longleftrightarrow \quad A \supseteq K_0.$$

Neither a singleton nor $\top$ contains K_0, so any such A belongs to $\mathcal{B}$.

First let a basic interval $J_{n,k}$ contain K_0. Since $\operatorname{diam}(K_0) = 1/2$, the estimate (5) implies

$$2^{1-n} \geq \frac{1}{2},$$

and hence $n \leq 2$. The endpoint conditions for $J_{n,k}$ to contain $[1/4, 3/4]$ are

$$\frac{k-1}{2^n} \leq \frac{1}{4}, \qquad \frac{k+1}{2^n} \geq \frac{3}{4}.$$

For $n = 1$ these inequalities force $k = 1$, so $J_{1,k} = J_{1,1} = X = \bot$. For $n = 2$ they force $k = 2$, so $J_{2,k} = J_{2,2} = K_0$.

Now let $A \in \mathcal{B}$ contain K_0 and write

$$A = J_{n_1,k_1} \cap \cdots \cap J_{n_r,k_r}.$$

Every basic interval occurring in this representation contains K_0 and is therefore either $\bot$ or K_0. Their intersection is consequently either $\bot$ or K_0. This proves (22). □

Proposition 5.6. *The complete lattice P is not continuous.*

Proof. If $A \ll K_0$, then $A \leq_P K_0$, because way-below implies below. By Theorem 5.5, the only possibilities are $A = \bot$ and $A = K_0$.

By Theorem 4.2,

$$\bot \ll K_0,$$

because $K_0 \subseteq \operatorname{int}_X \bot = X$. On the other hand,

$$K_0 \not\ll K_0,$$

because $K_0 \not\subseteq \operatorname{int}_X K_0$: the endpoints $1/4$ and $3/4$ do not belong to the relative interior of K_0. Hence

$$\Downarrow K_0 = \{\bot\}.$$

Its supremum is $\bot$, not K_0. Thus

$$K_0 \neq \bigvee \Downarrow K_0,$$

and P is not continuous. □

Remark 5.7. The proofs of Theorems 5.4 and 5.6 make the distinction between finite and one-point approximation explicit. A single element A satisfies $A \ll K$ precisely when $K \subseteq \operatorname{int}_X A$. At K_0, the sparse skeleton supplies no such approximant other than $\bot$. By contrast,

$$\mathcal{F}_m(K_0) = \{K_0, N_m(1/4), N_m(3/4)\}$$

uses one member for the interior of K_0 and two separate members for its endpoints. This finite branching is sufficient for quasicontinuity even though one-point approximation fails.

5.3. Failure of quasialgebraicity

Let ΣP denote P equipped with its Scott topology, and define

$$\eta : X \longrightarrow \Sigma P, \qquad \eta(x) = \{x\}. \tag{23}$$

Lemma 5.8. *The map η is continuous.*

Proof. Let $W \subseteq P$ be Scott open, and suppose that $\eta(x) = \{x\} \in W$. In the order of P, the sequence $(N_m(x))_{m\geq 1}$ is increasing and

$$\bigvee_{m\geq 1} N_m(x) = \{x\}.$$

Scott inaccessibility of W gives an index m such that $N_m(x) \in W$.

If $y \in N_m(x)$, then $N_m(x) \supseteq \{y\}$, and hence $N_m(x) \leq_P \{y\}$. Since W is an upper set, $\{y\} \in W$. Therefore

$$\operatorname{int}_X N_m(x) \subseteq \eta^{-1}(W).$$

The point x belongs to the open set on the left. Thus every point of $\eta^{-1}(W)$ has an open neighbourhood contained in $\eta^{-1}(W)$, and η is continuous. □

Lemma 5.9. *Let $\mathcal{F} \in P^{(<\omega)}$. If $\mathcal{F} \ll \mathcal{F}$, then*

$$\eta^{-1}(\uparrow_P \mathcal{F}) = \bigcup_{A\in\mathcal{F}} A \tag{24}$$

is a clopen subset of X.

Proof. For $x \in X$, formula (13) gives

$$\begin{aligned} x \in \eta^{-1}(\uparrow_P \mathcal{F}) &\iff \eta(x) \in \uparrow_P \mathcal{F} \\ &\iff \eta(x) \subseteq A \text{ for some } A \in \mathcal{F} \\ &\iff x \in \bigcup_{A\in\mathcal{F}} A. \end{aligned}$$

This proves (24). Every member of P is closed in X, so the finite union on the right is closed. By Theorem 2.1, $\mathcal{F} \ll \mathcal{F}$ implies that $\uparrow_P \mathcal{F}$ is Scott open. Its inverse image under the continuous map η is therefore open. Hence the trace is clopen. □

Proposition 5.10. *The complete lattice P is not quasialgebraic.*

Proof. Fix distinct points $x, y \in X$. Let $\mathcal{F} \in P^{(<\omega)}$ satisfy

$$\eta(x) \in \uparrow_P \mathcal{F}, \qquad \mathcal{F} \ll \mathcal{F}. \tag{25}$$

By Theorem 5.9, the set $\eta^{-1}(\uparrow_P \mathcal{F})$ is clopen in X. The first condition in (25) says that this trace contains x, so it is nonempty. Since $X = [0, 1]$ is connected, the trace must be all of X. In particular,

$$\eta(y) \in \uparrow_P \mathcal{F}.$$

Thus every Scott-open finitely generated upper set containing $\eta(x)$ also contains $\eta(y)$.

The family of such upper sets is nonempty: one has

$$\uparrow_P \{\bot\} = P,$$

which is Scott open, and hence $\{\bot\} \ll \{\bot\}$ by Theorem 2.1. Moreover, $\eta(x) \in P = \uparrow_P \{\bot\}$. Consequently,

$$\eta(y) \in \bigcap \left\{ \uparrow_P \mathcal{F} : \mathcal{F} \in P^{(<\omega)},\ \eta(x) \in \uparrow_P \mathcal{F},\ \mathcal{F} \ll \mathcal{F} \right\}.$$

On the other hand,

$$\uparrow_P \eta(x) = \{\eta(x), \top\}$$

does not contain $\eta(y)$. Hence the intersection required by quasialgebraicity at the element $\eta(x)$ is not equal to $\uparrow_P \eta(x)$, and P is not quasialgebraic. □

6. The counterexample and the interval map

Proposition 6.1. *Every chain in P is at most countable.*

Proof. The skeleton $\mathcal{B}$ is countable by Theorem 3.1. Distinct singleton subsets of X are incomparable under reverse inclusion: if $x \neq y$, then neither $\{x\} \supseteq \{y\}$ nor $\{y\} \supseteq \{x\}$. Thus a chain in P contains at most one singleton. All its remaining members lie in the countable set

$$\mathcal{B} \cup \{\top\}.$$

Therefore every chain is at most countable. □

The next proposition shows that the obstruction concerns interval chains and splitting, not the existence of an interval image.

Proposition 6.2. *Define $q : P \to [0,1]$ by*

$$q(A) = \begin{cases} \min A, & A \neq \top, \\ 1, & A = \top. \end{cases} \tag{26}$$

Then q is a Scott-continuous surjection.

Proof. If $A \neq \top$, then A is a nonempty compact subset of X, so $\min A$ exists. For every $t \in [0,1]$,

$$q(\{t\}) = t,$$

and hence q is surjective.

To prove monotonicity, suppose that $A \leq_P B$, equivalently $A \supseteq B$. If $B \neq \top$, then A is also nonempty and

$$\min A \leq \min B,$$

so $q(A) \leq q(B)$. If $B = \top$, then $q(B) = 1$, and the same inequality is immediate. Thus q is monotone.

Let $\mathcal{D} \subseteq P$ be directed, and put

$$S = \bigvee_P \mathcal{D} = \bigcap \mathcal{D}.$$

If $S = \top$, then Theorem 3.5 gives $\top \in \mathcal{D}$, and therefore

$$\sup_{D \in \mathcal{D}} q(D) = 1 = q(\top) = q(S).$$

Assume that $S \neq \top$, and define

$$\alpha = \sup_{D \in \mathcal{D}} q(D).$$

No member of $\mathcal{D}$ equals $\top$, since otherwise $S = \top$. Hence $q(D) = \min D$ for every $D \in \mathcal{D}$. Since $S \subseteq D$,

$$q(D) = \min D \leq \min S = q(S),$$

and consequently

$$\alpha \leq q(S). \tag{27}$$

Suppose that the inequality is strict. Choose $c \in \mathbb{R}$ with

$$\alpha < c < q(S) = \min S.$$

Then $S \subseteq (c, 1]$, which is open in the relative topology of X. By Theorem 3.4, there exists $D_0 \in \mathcal{D}$ such that

$$D_0 \subseteq (c, 1].$$

The set D_0 is nonempty because it contains S. Its minimum is therefore attained and lies in $(c, 1]$, so

$$q(D_0) = \min D_0 > c > \alpha,$$

contrary to the definition of α. Equality must hold in (27). Thus

$$q\left(\bigvee_P \mathcal{D}\right) = \sup_{D\in\mathcal{D}} q(D).$$

The map q preserves directed suprema and is Scott continuous. □

Proof of Theorem 1.3. The poset constructed in Section 3 is a complete lattice by Theorem 3.3. It is quasicontinuous by Theorem 5.4, not continuous by Theorem 5.6, and not quasialgebraic by Theorem 5.10. Every chain is at most countable by Theorem 6.1, and Theorem 6.2 supplies a Scott-continuous surjection onto $[0, 1]$.

If $C \subseteq P$ were a sub-dcpo order-isomorphic to $[0, 1]$, then C would be an uncountable chain in P, contradicting Theorem 6.1. If $[0, 1]$ were a Scott-continuous retract of P, there would be Scott-continuous maps

$$r : P \longrightarrow [0, 1], \qquad s : [0, 1] \longrightarrow P$$

with $r \circ s = \mathrm{id}_{[0,1]}$. The map s would be monotone and injective, so $s([0, 1])$ would again be an uncountable chain in P. This is impossible. □

Corollary 6.3. *Problems 1.1 and 1.2, equivalently Problems 1.14 and 1.15 of Xu, both have negative answers.*

7. A short note on splitting

The surjection q in Theorem 6.2 has no monotone section, and hence no Scott-continuous section: the image of a monotone injective map from $[0, 1]$ would be an uncountable chain. We record two elementary facts that isolate the additional compatibility needed for a general Scott-continuous surjection onto $[0, 1]$ to split. Throughout this section, $[0, 1]$ carries its usual order and its Scott topology.

Proposition 7.1. *Let $f : Q \to [0, 1]$ be a Scott-continuous surjection from a dcpo Q. Then f has a monotone section if and only if it has a Scott-continuous section.*

Proof. Every Scott-continuous section is monotone. Conversely, let $j : [0, 1] \to Q$ be monotone and satisfy $f \circ j = \mathrm{id}_{[0,1]}$. Define

$$\bar{j}(0) = j(0), \qquad \bar{j}(t) = \bigvee_{0\le r<t} j(r) \quad (0 < t \le 1). \tag{28}$$

The set under the supremum is a nonempty chain. For $t > 0$, Scott continuity of f gives

$$f(\bar{j}(t)) = \sup_{0\le r<t} f(j(r)) = \sup_{0\le r<t} r = t,$$

and $f(\bar{j}(0)) = 0$. Thus $\bar{j}$ is a section.

It remains to prove Scott continuity. Let $\mathcal{A} \subseteq [0, 1]$ be directed and put $t = \sup \mathcal{A}$. Monotonicity gives

$$\bigvee_{a\in\mathcal{A}} \bar{j}(a) \le \bar{j}(t).$$

If $t > 0$ and $r < t$, then some $a \in \mathcal{A}$ satisfies $r < a$; otherwise r would be an upper bound of $\mathcal{A}$. Hence

$$j(r) \le \bar{j}(a) \le \bigvee_{a\in\mathcal{A}} \bar{j}(a).$$

Taking the supremum over $r < t$ yields the reverse inequality. If $t = 0$, every member of $\mathcal{A}$ is 0, and equality is immediate. Therefore $\bar{j}$ preserves directed suprema and is Scott continuous. □

Let

$$\mathbb{D} = \left\{ \frac{k}{2^n} : n \geq 0,\ 0 \leq k \leq 2^n \right\}$$

be the set of dyadic rationals in $[0, 1]$.

Theorem 7.2. *Let $f : Q \to [0, 1]$ be a Scott-continuous surjection. The following conditions are equivalent:*

1. *f has a Scott-continuous section;*
2. *there exist elements*

$$x_d \in f^{-1}(d) \qquad (d \in \mathbb{D} \cap [0, 1))$$

such that

$$d < e \quad \Longrightarrow \quad x_d \leq x_e.$$

Proof. If $s : [0, 1] \to Q$ is a Scott-continuous section, then $x_d = s(d)$ has the required properties.

Conversely, suppose that an increasing exact dyadic family $(x_d)_{d \in \mathbb{D} \cap [0,1)}$ is given. Define

$$s(0) = x_0, \qquad s(t) = \bigvee_{\substack{d \in \mathbb{D} \\ d < t}} x_d \quad (0 < t \leq 1). \tag{29}$$

The family under the supremum is directed, and s is monotone. For $t > 0$, Scott continuity of f and density of $\mathbb{D}$ give

$$f(s(t)) = \sup_{\substack{d \in \mathbb{D} \\ d < t}} f(x_d) = \sup_{\substack{d \in \mathbb{D} \\ d < t}} d = t.$$

Also $f(s(0)) = 0$, so s is a section.

Let $\mathcal{A} \subseteq [0, 1]$ be directed and put $t = \sup \mathcal{A}$. Monotonicity gives

$$\bigvee_{a \in \mathcal{A}} s(a) \leq s(t).$$

If $t > 0$ and $d \in \mathbb{D}$ satisfies $d < t$, then some $a \in \mathcal{A}$ satisfies $d < a$, and therefore

$$x_d \leq s(a) \leq \bigvee_{a \in \mathcal{A}} s(a).$$

Taking the supremum over all dyadic $d < t$ yields the reverse inequality. The case $t = 0$ is immediate. Thus s preserves directed suprema and is Scott continuous. □

Remark 7.3. The criterion in Theorem 7.2 has the following finite form. Suppose that for each $d \in \mathbb{D} \cap [0, 1)$ a nonempty finite set $\mathcal{E}_d \subseteq f^{-1}(d)$ is prescribed, and that every finite dyadic grid

$$\mathbb{D}_n = \left\{ \frac{k}{2^n} : 0 \leq k < 2^n \right\}$$

admits an increasing choice $x_d \in \mathcal{E}_d$. The sets of such choices form the levels of a finitely branching tree under restriction. König's lemma gives an infinite branch, hence a compatible increasing family indexed by all dyadic rationals in $[0, 1)$. By Theorem 7.2, f then has a Scott-continuous section.

8. Concluding remarks

The counterexample is a complete lattice, so the negative answers do not arise from a lack of order completeness. They arise from the difference between finite and one-point approximation. The exact formula in Theorem 4.1 is the common mechanism: finite boundary choices yield quasicontinuity, the absence of sufficiently many one-point outer approximants yields noncontinuity, and connectedness prevents Scott-open finitely generated upper sets from separating the singleton layer. At the same time, Theorem 6.2 shows that a Scott-continuous interval image may exist without an interval chain or a compatible section.

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